

URBAN DEVELOPMENT AUTHORITY

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Office of the Executive Engineer, Circle I

NOTICE INVITING e-TENDER No. EE/WS/59/2026-27

Name of work: Electrical works for sewage treatment plant including motors and panels.

Estimated cost put to tender: Rs. 416,96,000/-

Earnest Money Deposit: Rs. 3,24,000/-

Period of completion: 9 months

Last date of submission: 13.11.2026 up to 15:00 hrs

PART A — INSTRUCTIONS TO BIDDERS

1. The tender documents may be downloaded from the e-procurement portal of the Government.
2. The bidder must be registered with the Department in the appropriate class.
3. Bids received after the due date and time shall not be considered.
4. Bidders shall have completed at least one similar work of value not less than 40 % of the estimated cost during the last seven years.
5. Conditional bids will be summarily rejected.
6. The bidder shall upload scanned copies of GST registration, PAN and audited balance sheets.
7. The successful bidder shall furnish a performance guarantee of 5 % of the contract value within 15 days.

COMMERCIAL TERMS AND CONDITIONS

1. Rates quoted shall be inclusive of all taxes, freight, insurance and transit risks.
2. The contractor shall indemnify the Department against all claims arising out of the work.
3. All disputes shall be subject to arbitration under the Arbitration and Conciliation

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Act, 1996.

4. Income tax and GST shall be deducted at source as applicable.

#### SCOPE OF WORK

This specification covers electrical works for sewage treatment plant including motors and panels complete in all respects.

#### SITE CONDITIONS

- i) Equipment is to be rated for a maximum ambient temperature of 50° C.
- ii) The relative humidity at site can go up to 95 pct.
- iii) The site is located at less than 1000 metre above MSL.

#### PART D — MAIN LT PANEL

- i) Distribution boards shall be of suitable dimensions for the equipment housed.
- ii) The main LT switchboard shall comply with IS/IEC 61439-1:2020 and IS/IEC 61439-2:2020.
- iii) The switchboard shall be of floor standing cubicle construction using CRCA sheet steel 2 m.m. thick.
- iv) The switchboard shall have a rated operational voltage of 415 V AC, 3 phase 4 wire, 50 Hertz.
- v) The main busbar shall be electrolytic grade aluminium with a continuous rating of 1600 Amperes.
- vi) The short time withstand current of the busbars shall be 50 kA (rms) for one second.
- vii) The panel enclosure is to have a degree of protection of IP54.
- viii) Internal separation of the switchboard shall be to Form 3b.
- ix) Each incomer shall be a 1600 Amperes, four pole, motorised air circuit breaker complying with IS/IEC 60947-2:2003.
- x) Outgoing feeders shall be controlled by MCCBs having an ultimate breaking capacity of

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36 kA (rms).

xi) Each incomer shall be provided with a class 0.5 accuracy multifunction meter.

#### GUARANTEED TECHNICAL PARTICULARS — LT PANEL

Rated operational voltage: 415 V AC

Main busbar rated current: 1600 Amperes

Rated short-time withstand current: 50 kA (rms) for 1 s

Enclosure degree of protection: IP54

Make and type: (to be furnished by the bidder)

#### SECTION 5: DRIVE MOTORS

i) Squirrel cage induction motors complying with IS 325:1996 shall be supplied.

ii) Motor rated output shall be 37 k.W..

iii) Motors are to operate on 415 V AC  $\pm$  10 pct, 50 Hertz  $\pm$  5 pct, three phase supply.

iv) Motors shall meet efficiency class IE3 of IS 12615:2018.

v) The motor enclosure shall have IP55 degree of protection.

vi) Insulation shall be class F, with temperature rise restricted to class B, for continuous S1 duty.

vii) The motor synchronous speed shall be 1500 rev/min.

#### SECTION 6: LT POWER AND CONTROL CABLES

i) LT power cables shall be aluminium conductor, XLPE insulated, armoured, PVC sheathed, 1.1 kV grade, complying with IS 7098 (Part 1):1988.

ii) Maximum continuous conductor temperature of the cables shall be 90° C.

iii) Copper conductor control cables of 2.5 Sq mm, PVC insulated, 1.1 kV grade shall comply with IS 1554 (Part 1).

iv) Class 2 stranded conductors complying with IS 8130 shall be used.

v) Cables shall be run in RCC trenches or on perforated trays as per drawings.

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- vi) Double compression glands and crimped lugs shall be used for all cable terminations.
- vii) The maker's name, size, voltage grade and year of manufacture shall be marked on every drum.

#### 7 TESTS AND INSPECTION

- i) Each assembly shall undergo routine verification as laid down in IS/IEC 61439-1.
- ii) The bidder shall furnish type test certificates of the offered panel design issued by a NABL accredited laboratory.
- iii) Each motor shall be subjected to routine tests as per IS 4029.
- iv) All motors shall carry the ISI mark.
- v) Testing of the cables shall be done as per IS 10810.
- vi) Each consignment of cables shall be accompanied by routine test certificates.
- vii) Before despatch, the Engineer-in-charge will inspect the equipment at the maker's works.

#### 8 DOCUMENTATION, WARRANTY AND TRAINING

- i) Detailed shop drawings are to be submitted by the contractor for approval not later than 21 calendar days after the award of work.
- ii) On completion the contractor shall hand over three copies of the O&M manuals.
- iii) The contractor shall guarantee the whole installation for 12 months after handing over against faulty design, materials and workmanship.
- iv) Operation and routine maintenance training shall be given by the contractor to the Department's operators.

#### SCHEDULE OF QUANTITIES

Sl. No. | Description | Unit | Qty

- 1 | Main LT panel as per specification | Set | 1
- 2 | Sub-distribution board, 12 way TPN | No. | 8

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3 | Squirrel cage induction motor 37 kW, 1500 rpm | No. | 3

4 | 3.5C x 300 sq mm XLPE armoured cable | m | 350

5 | Cable tray 300 mm wide | m | 120

APPROVED MAKES OF MATERIALS

LT panel builder: as per approved list; ACB/MCCB: L&T / Siemens / Schneider / ABB

Motors: ABB / Siemens / Crompton Greaves / Kirloskar

Cables: Polycab / KEI / Havells / RR Kabel

Note: Drawings are indicative only and shall be verified at site.

Note: In case of any discrepancy between drawings and specifications, the decision of the Engineer-in-charge shall be final.

Signature of Bidder with seal

Executive Engineer

Signature of Bidder

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